LMS Algorithm Derivatives used in Real-Time Filtering of ECG Signals: A Study Case on Performance Evaluation

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Abstract— **The paper presents two derivatives of the Least Mean Square (LMS) algorithm, which can be utilized in real-time processing of the electrocardiographic (ECG) signals: (i) the Normalized LMS (NLMS), and (ii) Sign based LMS (S-LMS) algorithms. The algorithms are suitable for implementation on Digital Signal Processors (DSPs) or on Field Programmable Gates Arrays (FPGAs), and the practical implementations of their equations are presented. The implementation of the iterative steps can be done in an easy manner with low computational complexity with good effect on the filtering process, for both types of perturbations, electronic noises and power line interference signals, which can occur on ECG signals.**

Keywords - adaptive filter, ECG signal processing, real-time signal processing, LMS algorithm.

I. INTRODUCTION

Before doing any kind of analysis of the ECG signal, it is firstly necessary to be done a filtering operation to remove all unwanted signals that can disrupt the ECG waveform [1]. A lot of papers have been written over the years about cleaning methods of the ECG signal. With the growth of power of computers, researchers have redirected their interest to applying filtering solutions for ECG signals, by using intelligent algorithms such as adaptive filters, wavelets, neural networks or fuzzy systems. Nowadays, the analog or digital integrated circuits are developed by using advanced technologies, and for this reasons, their cost is in a permanent decreasing. Modern electrocardiographs includes an mixt analog and digital techniques for perturbation suppression [2],[3]. The aim of the paper is to present LMS algorithms derivatives, as a practical solution which can be adopted for the adaptive filtering of the ECG signals.

After the introductive section, the paper is organized as it follows. Section II presents the problem of noises and interfering signals, which can affect the ECG signal waveform. After this, all mathematical equations of LMS derivatives, for practical approach on an adaptive filter structure are described in Section III. Simulation results were performed, and they are presented in Section IV. Last section concludes the paper.

II. NOISES AND INTERFERENCES: THE PERTURBATIONS OF THE ECG SIGNALS

The digital techniques to suppress the perturbation from ECG signals are preferred of being studied by most researchers [4]-[8]. Also, by using digital techniques, it can be removed baseline variation of ECG signal on his acquisition [9]. But this variation of ECG signal baseline can't be considered as a major source of perturbations. In [1] and [10], two types of perturbations, electronic noises and interferences, as major sources of unwanted signals which can produce alterations on waveform of ECG signal, are presented. The electronic noises could be produced by connection paths, or by the silicon components from the acquisition circuitry of ECG signal. An example of a random noise signal type, which has been appeared on a real ECG signal, is shown in Fig. 1. Signals interference types are produced by electrical power source through capacitive coupling or by the conduction. An example of an interference signal of 50 Hz frequency, which may contains harmonics which overlaps on real ECG signal, is shown on Fig. 2.

Therefore, first of all, by knowing the type of disturbing signal, which may affect the waveform of the ECG signal, it will be possible for further to search and apply an optimal solution, in terms of the filtering process. It is known that, the electronic noise or the 50 Hz interference will not always be overlapped in spectral composition of ECG signal [1]. So, for this reason, in time, different approaches for adaptive filter structures for each type of noise were proposed by researchers.

One important issue of the adaptive filtering methods is the problem of signal's correlations [11], regarding to reference signal source of perturbation and overlapped perturbation on ECG signals. After this short introduction to the perturbations problem, regarding adaptive filtering of the ECG signals, the readers can have a good idea about it, in a real way of thinking. In the followings, few recent scientific works, with relevance on adaptive filtering of ECG signal, will be discussed. While reading this paper, readers are also encouraged to follow and check the references from this paper.

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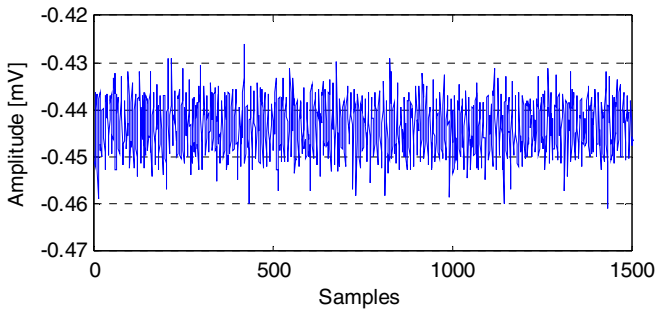


Figure 1. Electronic noise overlapped on ECG signal.

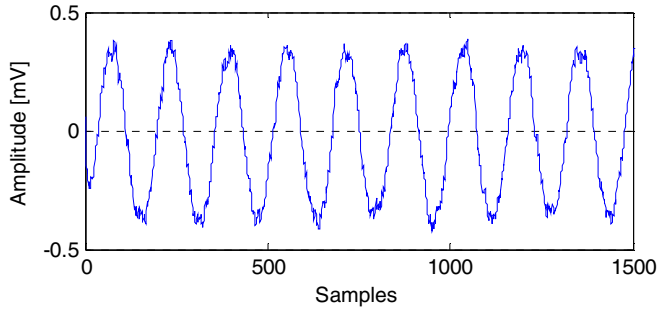


Figure 2. Power line interfering signal.

In this way, a good understanding state of the adaptive filtering principles could be achieved.

In [11], for the first time in the history, of the adaptive filtering of ECG signal, the authors describe one possible solution, very well argued. Here, the LMS algorithm is proposed, and tests on removing the 50/60 Hz power line interfering signals from ECG signals, are performed and presented. From this paper until today, all scientific works that presents an algorithm based on adaptive method, which is applied to eliminate the 50/60 Hz power line interfering signals from the ECG waveforms, have this approach at the basic. The adaptive filter structure proposed by Widrow, which can be used to eliminate the perturbation component of 50/60 Hz overlapped on an ECG signal, is presented on Fig. 3. Here, $e[k]$ represents the error signal, given by the equation (1):

$$e[k] = d[k] - y[k] \tag{1}$$

The noisy ECG signal is represented as $d[k]$ and the filter output is represented by $y[k]$, given by the following eq. (2):

$$y[k] = w_1[k] * x_1[k] + w_2[k] * x_2[k] \tag{2}$$

where, the filter coefficients are represented by the $w_1[k]$ and $w_2[k]$, the reference signal by the $x_1[k]$ and $x_2[k]$, which contain the 50/60 Hz interference samples, at the k moment of time, and * represents the convolution operator. For each input sample the adaptive algorithm will recalculate and update in a recursive manner the values of $w_1[k]$ and $w_2[k]$. Another very used structure of an adaptive filter, which is presented in Fig. 4, can be applied for electronic noise removal from ECG signals. Here, the filter's output $y[k]$ is given by the following formula (3):

$$y[k] = \sum_{i=1}^{n} w_i[k] * x_i[k] \tag{3}$$

where, n represents the length of the Finite Impulse Response (FIR) filter type and z^{-1} represents units delay. The adaptive filter structure, which is presented in Fig. 3, is assimilated on literature as an adaptive notch filter type. By a notch filter type, it can be understood a band stop filter. On the other hand, the filter structure shown in Fig. 4, shall be treated as a type of low-pass FIR filter, with the role of eliminating from of the ECG signal spectrum, unwanted electronic noises of high frequency.

With the time, since the first adaptive algorithm and filter were proposed by Widrow in 1975, until today, many versions of practical implementation have been proposed by the scientific community in the field of ECG filtering. Therefore in [12]–[35], by using a complex mathematical description, or by having intentions to show a possible practical implementation, different algorithms and structures of filters, have been proposed. From all presented issues until here, a short conclusion can be suggested. The adaptive filtering of the ECG signal could be developed and applied with best results into real electrocardiographs as an alternative to other solutions.

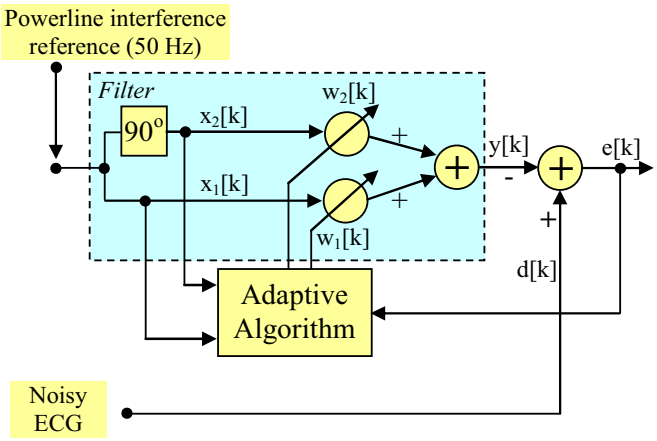


Figure 3. Adaptive filter principle used to eliminate 50 Hz component.

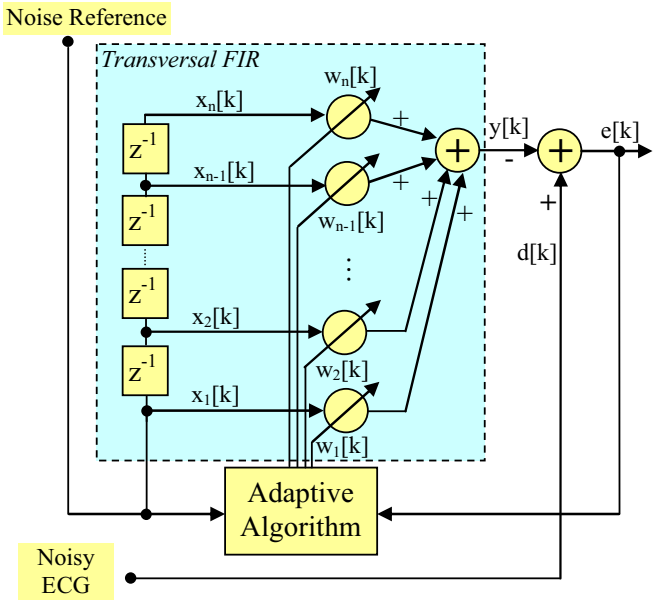


Figure 4. Adaptive filter principle used to eliminate electronic noise component.

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III. ALGORITHMS DESCRIPTION

Until now, the filtering mechanisms that are based on structures of the adaptive filters have been presented. Starting from these structures of adaptive filters, in the followings, the LMS algorithm with his derivatives, which can be applied on practice, will be presented. Until now, in various works such as [36]–[42], the authors have tried to apply the following adaptive algorithms for presented structures of adaptive filters: LMS, NLMS, Sign-Data LMS, Sign-Error LMS and Sign-Sign LMS. Their intention was to promote different adaptive algorithms for practical approach, and to present also results for denoising of ECG signal.

In Table I, the necessary resources allocation (e.g. multiplications, additions, divisions or memory allocations) for practical implementations, indifferently if the adaptive filters are implemented on DSP processors or FPGA circuits, which are required for filter and each algorithm, are presented. According to Table I, in Table II the mathematical description of each type of adaptive algorithm used for coefficients updating of the adaptive filters are presented. From this table, we can see that all derivatives of LMS algorithms are based on the same mathematical structure of the LMS algorithm equation, which are used for coefficients updating at the $k+1$ iteration step. It should be noted that the allocation of resources which are needed to implement the adaptive algorithms either on DSP processors or on FPGA circuits is identical.

In Table III, the execution steps for the LMS algorithm, which can run on DSP processors or on FPGA circuits, are shown. The same mode of execution of the LMS algorithm steps is applied also to the rest of the presented algorithms above.

TABLE I. RESOURCES ALLOCATION.

Necessary operations to be performed		Mult. (*)	Add. (+)	Div. (/)	Mem. access
LMS	***Filter***	*n*	*n*	-	*2n*
	Algorithm	*2n+1*	*2n*	-	*2n+4*
NLMS	***Filter***	*n*	*n*	-	*2n*
	Algorithm	*2n+4*	*2n+2*	*1*	*2n+7*
Sign-Data LMS	***Filter***	*n*	*n*	-	*2n*
	Algorithm	*2n+1*	*2n*	-	*2n+4*
Sign Error LMS	***Filter***	*n*	*n*	-	*2n*
	Algorithm	*2n+1*	*2n*	-	*2n+4*
Sign-Sign LMS	***Filter***	*n*	*n*	-	*2n*
	Algorithm	*2n+1*	*2n*	-	*2n+4*

Note: n – maximum filter length.

TABLE II. PRACTICAL IMPLEMENTATION OF ALGORITHMS.

Algorithm	Equation
LMS	$w_n[\mathrm{k}+1] = w_n[\mathrm{k}] + \mu \cdot e[\mathrm{k}] \cdot x_n[\mathrm{k}]$
NLMS	$w_n[\mathrm{k}+1] = w_n[\mathrm{k}] + \frac{\mu}{\lVert x[n] \cdot x[n] \rVert} \cdot x_n[\mathrm{k}] \cdot e[\mathrm{k}]$
Sign-Data LMS	$w_n[\mathrm{k}+1] = w_n[\mathrm{k}] + \mu \cdot e[\mathrm{k}] \cdot \mathrm{sign}(x_n[\mathrm{k}])$
Sign-Error LMS	$w_n[\mathrm{k}+1] = w_n[\mathrm{k}] + \mu \cdot \mathrm{sign}(e[\mathrm{k}]) \cdot x_n[\mathrm{k}]$
Sing-Sign LMS	$w_n[\mathrm{k}+1] = w_n[\mathrm{k}] + \mu \cdot \mathrm{sign}(e[\mathrm{k}]) \cdot \mathrm{sign}(x_n[\mathrm{k}])$

Note: n – maximum filter length, $n=1...n$; k – the sample at the k moment of time.

TABLE III. MODE OF OPERATION OF THE LMS ALGORITHM.

Algorithm steps:	
1.	The cycle counter variable is initialized: i=0, and it is assigned an initial value for the vector of coefficients: $w(n) = [0\ 0\ 0...0]^{\mathrm{T}}$, and the μ constant is chosen properly.
2.	Calculate the current output of the filter: $y[k] = \sum_{i=1}^{n} w_i[k] * x_i[k]$
3.	Determine the current sample of the error signal: $e[k] = d[k] - y[k]$
4.	Calculate the weight vector for the next step: $w_n[\mathrm{k}+1] = w_n[\mathrm{k}] + \mu \cdot e[\mathrm{k}] \cdot x_n[\mathrm{k}]$
5.	Counter variable is incremented: i=i+1; $i = 0,...,n$ and executes a jump to step 2.

The LMS algorithm and also all its derivatives are based on the Steepest Descent (SD) method [11], as a mechanism for convergence criteria. According to this method, the algorithms will converge to optimal solution and will minimize the Mean Square Error (MSE) in a finite number of iterations. This MSE parameter can be considered as a metric of performance in evaluating of the adaptive algorithms. According to this metric, the test simulations could be created in order to track for all algorithms, their performances.

Besides this parameter MSE, it can be tracked also the evolution of the adaptive algorithms according to the adaptive filter length, number of iterations, the variation of convergence parameter μ (miu) and stability in time of algorithms. In the following section, few simulation tests for the presented performance criteria, are presented and discussed.

IV. SIMULATION RESULTS AND DISCUSSIONS

In Fig. 5, the effect produced by the variation of the adaptation parameter μ on the MSE convergence criteria for the LMS algorithm, is presented. For parameter μ, the test values from 0.5 to 0.0001 were given.

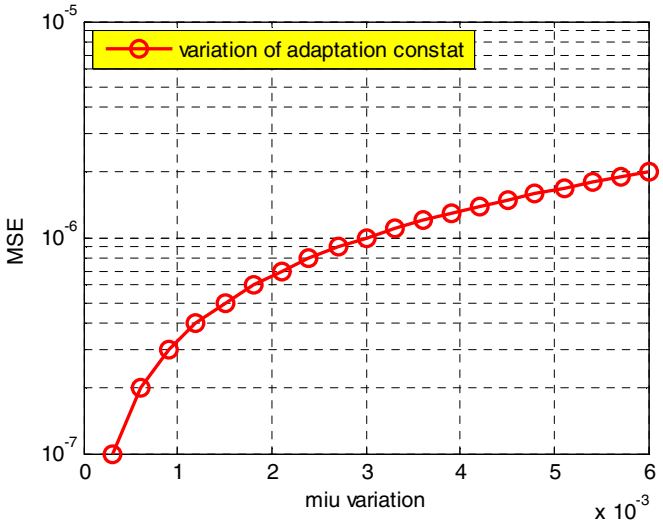


Figure 5. The effect produced of variation of the adaptation parameter μ on MSE for LMS algorithm.

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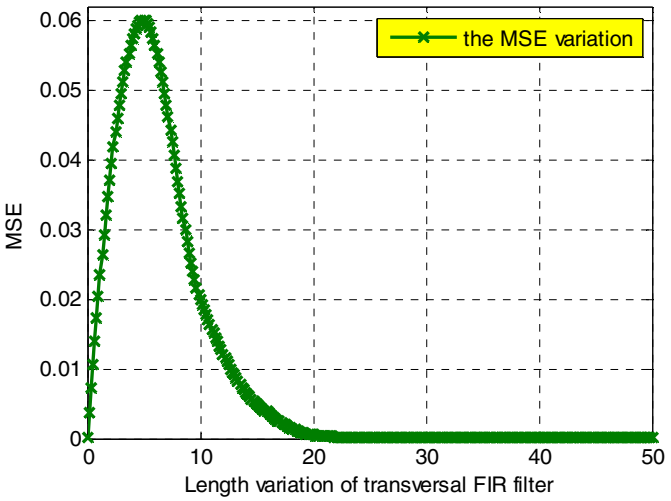


Figure 6. The produced effect for MSE by variation of the filter length for the LMS algorithm.

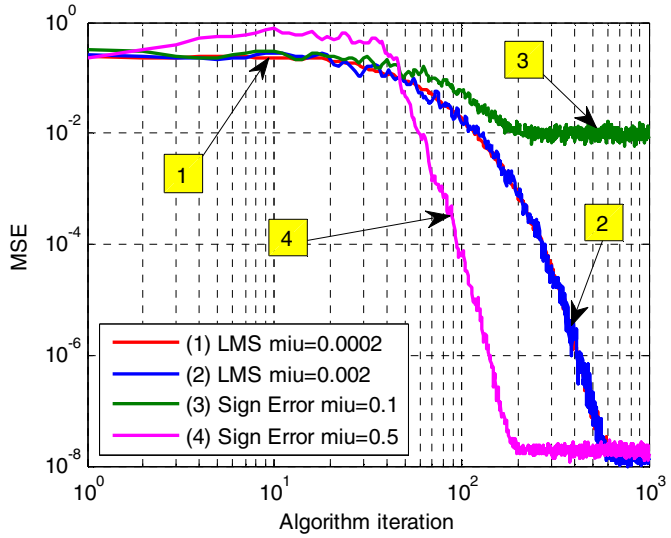


Figure 7. LMS and Sign-Error LMS algorithms learning curves for different values of miu

All the simulation tests were conducted for the adaptive filter structure presented in Fig. 4. The adaptive filter structure which is depicted in Fig. 3 represents a particularity of the adaptive filter structure presented in Fig. 4. All simulations were performed on a standard waveform of ECG signal [1].

The MATLAB software was used to perform all the algorithms implementation which was tested. Also the LabVIEW software can be used as an alternative.

One important test is regarding to the effect produced for MSE by varying of the filter length and the obtained results for the LMS algorithm are presented in Fig. 6. From these results, it can be seen that for low filter lengths, the values of MSE are greater than for long filter lengths.

Other tests which were performed are based on MSE variation, depending on the number of algorithm iterations for the adaptive filters.

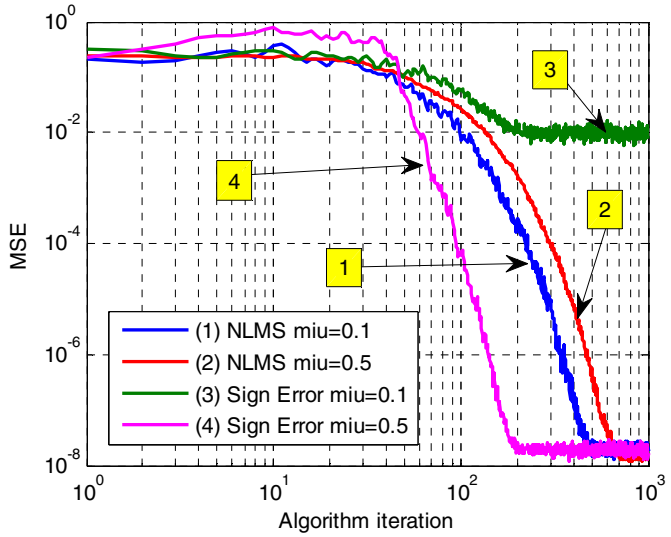


Figure 8. NLMS and Sign-Error LMS algorithms learning curves for different values of miu

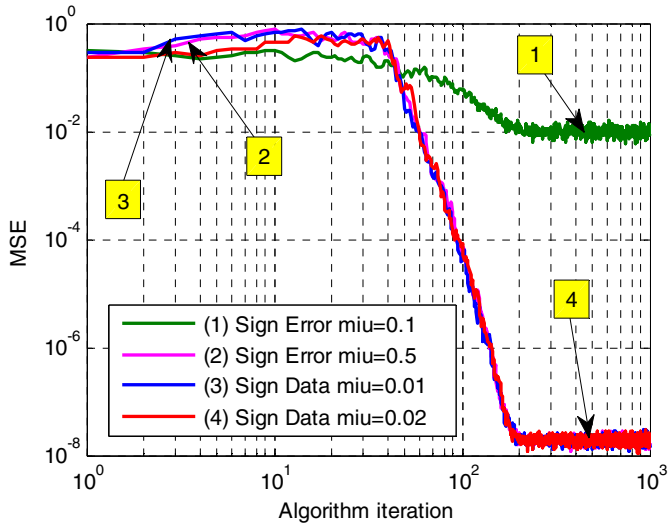


Figure 9. Sign-Error and Sign-Data LMS algorithms learning curves for different values of miu

The comparison of the results for the LMS and Sign-Error LMS algorithms of the learning curves for different values of μ (miu), are presented in Fig. 7. From this figure, it can be seen that for the adaptation constant μ=0.5 of the Sign-Error LMS algorithm, it can be produced approximately the same behavior as the LMS algorithm.

For the same Sign-Error LMS algorithm, in another simulation case, a poor convergence is obtained for the μ=0.1. In the following, other results for the simulations regarding to the NLMS and Sign-Error LMS algorithms are presented in Fig. 8.

The values for μ for the Sign-Error LMS algorithm were kept the same, as in previous experiment. In order to create comparisons for the behavior of the NLMS algorithm, two values for μ (Fig. 8) were suggested.

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The convergence of the NLMS algorithm is faster for μ=0.1 than for μ=0.5, but for the last one, the steady state error is less in comparison, if the μ is chosen 0.1.

In Fig. 9, the learning curves for the Sign-Error LMS and Sign-Data LMS algorithms are presented. From this figure, it can be seen that for three simulations the results coincide in a certain way. From the presented results, a short conclusion can be proposed. The behaviors of the MSE for the same values of μ, which are applied on different adaptive algorithms, will not be the same. After performing simulations, the best obtained results, according to the presented metrics of performances, can be utilized to be applied for practical implementation of adaptive filters which can be used for ECG signal processing.

All performances of the all adaptive algorithms could be affected on practical implementations, by the finite word length of the samples. Also, the convergence speed for all adaptive algorithms depends on the frequency sampling of the real signals which are applied to the adaptive filters.

In papers [43]–[45], several performance metrics, which are proposed to be used, to evaluate the digital algorithms used in filtering of the ECG signal are also presented and discussed.

V. CONCLUSIONS

In this paper, few issues of the adaptive filtering techniques, which are applied for denoising of the ECG signal, were presented. A study case on performance evaluation of the adaptive algorithm was described. Few practical aspects for the case of real implementations for the adaptive filters were suggested.

The specialty literature is vast in the field of adaptive filtering of ECG signals, and there are few papers (see the references pointed in this paper) which describe the aspects for a possible practical application of these adaptive algorithms for ECG signal processing.

Usually most scientific published papers presents the adaptive filtering with a hard mathematical model which makes them sometimes difficult to understand. Sometimes in those scientific papers, the issues regarding to the signals theory are presented in the detriment of the digital signal processing issues.

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